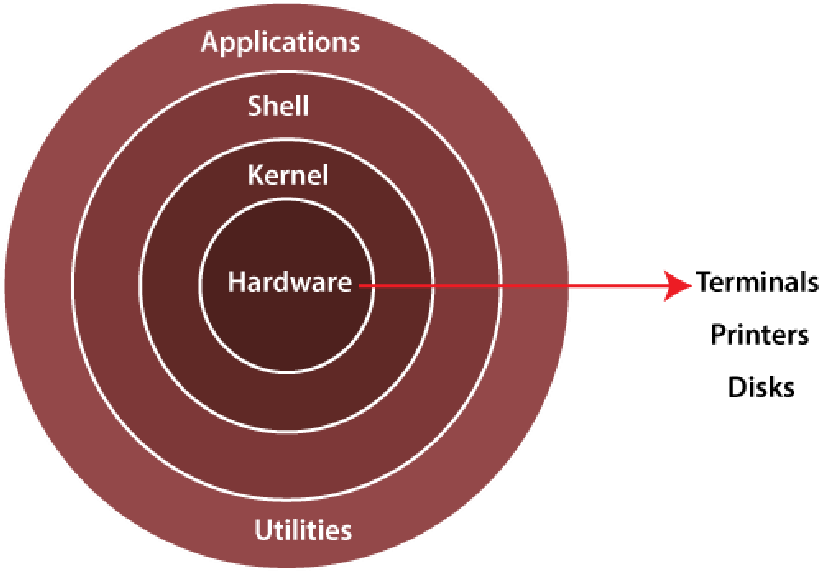
Architecture of Linux system



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The Architecture of a Linux System Consists of the following layers.

- **Hardware Layer** :- Hardware consists of all peripheral device (RAM/HDD/CPU etc).
- **Kernel** :- It is a core component of Operating System, interacts directly with hardware and software.
  It is also known as Heart of Linux.
- **Shell** :- A Shell is a interpreter that acts as an interface between the user and operating system.
  The Shell takes commands from the users and executes kernels functions.
  Popular shells include BASH (Bourne Again Shell), Zsh (Z Shell) and Fish (Friendly Interactive Shell).

- **Utilities** :- Utilities refers to small software programs or commands that perform specific task or function.
  Linux utilities cover a wide range of functionalities, including file manipulation (e.g. copy, move, delete),
  Text processing (e.g. search, replace, format).

2. Features of Linux

- **Open Source** – Free to use, modify, and distribute.
- **Multi-user** – Multiple users can use the system simultaneously.
- **Multitasking** – Can run multiple programs at once.
- **Portability** – Runs on different hardware platforms.
- **Security** – User permissions, firewalls, SELinux, etc.
- **Stability & Reliability** – Used for mission-critical servers.
- **Shell & Command Line Interface (CLI)** – Powerful control for users.

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3. Components of Linux

1. **Kernel** – The core part of Linux that manages hardware, memory, processes, and devices.
2. **Shell** – Command interpreter (e.g., Bash, Zsh).
3. **File System** – Organizes data (ext4, xfs, btrfs).
4. **System Libraries** – Standard functions for applications.
5. **Utilities** – Commands & tools (ls, cp, mv, ps, grep).

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4. Linux File System Hierarchy (FHS)

- / → Root directory
- /bin → Essential user binaries
- /etc → Configuration files

- /home → User home directories
- /var → Variable files (logs, mail, spool)
- /tmp → Temporary files
- /usr → User programs and utilities
- /boot → Boot loader files & kernel
- /dev → Device files
- /proc → Kernel & process info (virtual FS)

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5. Basic Linux Commands

- **File/Directory Handling**: ls, cd, pwd, mkdir, rmdir
- **File Operations**: cp, mv, rm, touch, cat, nano/vi
- **Permissions**: chmod, chown, umask
- **Process Management**: ps, top, kill, jobs, fg, bg
- **System Info**: uname, df, du, free, uptime
- **Networking**: ping, ifconfig/ip, netstat, scp, ssh

COMMANDS :-

$date :- this command shows date with time.

$pwd :- present working directory.

$date +%d :- date 15.

$date +%m :- month

$date +%Y :- year.

$whoami :- it shows the current login user.

$who :- **Tell us which user is using which terminal on which date and time**

$uptime :- cpu utilization/load average.

$history :- history command shows the history on server.

What is a Command?

Instructions given to OS.

$ man date

Man means manual or help command.

$uname -r :- kernal version.

-a :- you will get the information of whole system

$touch file1

$touch file1, file2, file3.

$ ls :- list the names of files / directory.

$echo :- is used to print message.

$clear :- command is used to clear your screen.

$alias :- It is a nickname of the command.

If you want to use any shortcut so you have to use alias command and then the command name.

$id :- it gives you user details.

$ifconfig :- is used to know the IP address your system

$uptime :- is used to load average of a system.

$w :- This command gives you time details of user active/login.

$user :- It shows your current logined user.

$hostname :- its shows full name of your system.

$hostname -s :- hostname.

" -i :- it shows Ip address.

$ls -a :- its show me the hidden file and directory.

$cal :- it shows calendar .

$cal 2 2023.

$cal 8 1947.

uname :- this command tell us the OS name

uname -i :- it will show the architecture.

Uname -a :- It will give detail information it will show you the distribution kernel version , architecture.

#w :- it gives more detail than who command.

#users:- This command tell us currently which users are login in our system.

#hostname :- hostname is also a identity of our system.

Or to get the system hostname.

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6. Linux User Management

- Users & groups control access.
- **Commands**:
  - useradd, passwd, usermod, userdel
  - groupadd, groups, gpasswd

  File permissions:
  - **r** (read), **w** (write), **x** (execute).
  - Ownership: **User, Group, Others**.

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7. Linux Process Management

- **Process** = Running program.
- Process types: Foreground & Background.
- Commands:
    - ps – show running processes
    - top/htop – real-time monitoring
    - kill – terminate a process
    - nice/renice – set process priority

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8. Linux Distributions

Popular distros:

- **Enterprise**: Red Hat Enterprise Linux (RHEL), SUSE Linux Enterprise
- **Community**: Ubuntu, Debian, Fedora, CentOS, Arch Linux
- **Specialized**: Kali Linux (security), Android (mobile)

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9. Linux System Administration Basics

- Package Management:
    - **RHEL/CentOS** → yum, dnf, rpm
- Service Management:
    - systemctl start/stop/restart/status <service>
- Logs: Stored in /var/log/
- Scheduling: cron, at

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10. Advantages of Linux

- Free & Open Source
- Highly Secure & Stable
- Performance-efficient

- Large Community Support
- Ideal for servers, cloud, and development